PMP CERTIFICATION COURSE

Student Reference Resource Guide

Section 6

40 Processes, Inputs, Tools & Outputs

All 15 Lectures Covered

PMBOK Guide 8th Edition	2026 ECO Aligned

Section 6 Overview

Section 6 examines all 40 project management processes defined in the PMBOK Guide. These processes provide the procedural backbone for managing any project — they define what work is performed, when it is performed, and what is produced at each stage.

The 40 processes are distributed across five Process Groups (Initiating, Planning, Executing, Monitoring & Controlling, and Closing) and ten legacy Knowledge Areas (Integration, Scope, Schedule, Cost, Quality, Resources, Communications, Risk, Procurement, and Stakeholder Management). PMBOK 8 aligns these processes to seven Performance Domains, but knowledge of Knowledge Areas remains essential for ITTO-based exam questions.

Each process is defined by its Inputs, Tools & Techniques, and Outputs (ITTOs). Mastering ITTOs enables you to answer situational exam questions by reasoning about what a process needs, what it does, and what it produces — rather than guessing.

Process Group	# Processes	Key Processes
Initiating	2	Develop Project Charter; Identify Stakeholders
Planning	24	All subsidiary plans + scope, schedule, cost, quality, resource, communications, risk, procurement, stakeholder management processes
Executing	10	Direct & Manage Work; Manage Knowledge; Manage Quality; Acquire/Develop/Manage Team; Manage Communications; Implement Risk Responses; Conduct Procurements; Manage Stakeholder Engagement
Mon. & Controlling	12	Monitor & Control Work; Integrated Change Control; Validate/Control Scope; Control Schedule/Costs/Quality/Resources/Comms/Risks/Procurements; Monitor Stakeholder Engagement
Closing	1	Close Project or Phase

Why ITTOs Matter for the Exam: The Process Domain accounts for 41% of PMP exam questions (2026 ECO). ITTO knowledge is tested both directly ("What is an output of...") and situationally ("The project manager notices X — which process produces this information?"). Understanding process flow — how outputs become inputs — is more powerful than rote memorisation.

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Meet Sarah and the Riverside Medical Center — Your Project Begins

Overview

This opening lecture introduces the Riverside Medical Center project — a $50 million, three-year hospital build that serves as the running case study for all 40 processes. Rather than presenting processes as abstract theory, the course grounds each one in concrete decisions Sarah and her team must make.

The lecture establishes the project's scale: 200 beds, five operating theatres, an emergency department, a radiology wing, and full IT infrastructure. This complexity demands every PMBOK 8 process to be applied correctly.

Students learn why the case-study approach accelerates exam readiness: seeing the same project evolve through initiating, planning, executing, monitoring, and closing makes process sequencing intuitive rather than memorised.

Key Concepts

- The 40 processes are grouped across the five Process Groups: Initiating (2), Planning (24), Executing (10), Monitoring & Controlling (12), Closing (1) — these numbers don't add to 40 because PMBOK 8 restructures some. Know that PMBOK 8 aligns processes to 7 Performance Domains.
- A project sponsor authorises the project and provides resources. For Riverside Medical Center the city government acts as sponsor.
- The project charter is the founding document that formally authorises the project and names the project manager.
- Stakeholders are identified as early as initiation — delay risks misaligned expectations that compound through the project lifecycle.
- The case study approach mirrors how the PMP exam tests situational judgement: you must apply the right process in the right context, not recall definitions in isolation.
- PMBOK 8 is principles-based, but the 40 processes provide the procedural backbone that connects those principles to daily project work.
- A project manager's authority level is defined in the project charter — this is especially important in matrix organisations like a city hospital build.
- Inputs, Tools & Techniques, and Outputs (ITTOs) are the anatomy of every process. Understanding them proves to the exam that you know what goes in, what you do, and what comes out.
- The hospital project involves multiple performing organisations: construction, IT, medical equipment, and the city's PMO — making integration management critical from day one.

Key Terms & Definitions

Term	Definition
Project Charter	Formal document that authorises the project, names the PM, and grants authority to apply organisational resources.
Sponsor	Person or group who provides financial and political support for the project and authorises the charter.
ITTO	Inputs, Tools & Techniques, Outputs — the structural elements that define what each process consumes and produces.
Process Group	Logical grouping of processes by phase: Initiating, Planning, Executing, Monitoring & Controlling, Closing.
Performance Domain	PMBOK 8 grouping of related activities critical to delivering project outcomes (e.g., Stakeholders, Team, Delivery).
Stakeholder	Any individual, group, or organisation that may affect, be affected by, or perceive itself to be affected by the project.

✦ EXAM TIPS

- The exam tests application, not memorisation. When a question describes a situation, identify which process group the project is in before choosing an answer.
- Know the two Initiating processes cold: Develop Project Charter and Identify Stakeholders. Any question about 'project beginning' almost always maps to one of these.
- PMBOK 8 no longer presents Knowledge Areas as the primary structure — but the 40 processes still exist. Map them to both Performance Domains (for the exam) and legacy Knowledge Areas (for ITTO recall).
- The Riverside Medical Center scenario mirrors PMP exam situational questions. Practise asking: 'What process is Sarah in, and what should she do next?'
- The project charter authorises the PM — if a question asks who should resolve a resource conflict or approve scope, the answer often traces back to the charter.

✦ KNOWLEDGE CHECK

1. Sarah has just been named project manager for the Riverside Medical Center. The city has provided a document defining the project objectives, budget, and her authority. What is this document called, and which process produced it?
2. A question states: "The team is in the earliest stage of a project and needs to formally authorise the work." Which Process Group are they in, and which process should they execute first?
3. Why does PMBOK 8 use Performance Domains instead of Knowledge Areas as its primary structure, and how does this affect how you study the 40 processes?

4. Sarah identifies six organisations that will contribute to the hospital build. At what point in the project should she formally identify these stakeholders, and why does timing matter?

The Master Map — All 40 Processes Across 7 Performance Domains

Overview

Lecture 2 is the master map: all 40 processes laid out across process groups and performance domains so students can see the entire project management system in one view. This helicopter perspective is essential before diving into individual processes.

The lecture explains how the five Process Groups describe the flow of work across a project's life, while the performance domains describe the areas of focus. Each of the 40 processes sits at an intersection — it belongs to one process group and relates to one or more performance domains.

Students learn the exam-critical insight: processes are not strictly sequential. Multiple planning processes run in parallel, and monitoring processes run continuously alongside execution. Understanding flow prevents the mistake of treating project management as a rigid checklist.

Key Concepts

- Initiating (2 processes): Develop Project Charter, Identify Stakeholders. These formally open the project and establish the foundational stakeholder register.
- Planning (24 processes): The largest group. Covers scope, schedule, cost, quality, resources, communications, risk, procurement, and stakeholder engagement plans.
- Executing (10 processes): Where deliverables are produced. Includes directing work, quality assurance, team development, and stakeholder engagement.
- Monitoring & Controlling (12 processes): Tracks performance, controls changes, and manages variances. Runs simultaneously with Executing.
- Closing (1 process): Close Project or Phase — finalises all activities, releases resources, documents lessons learned, and formally closes the project.
- The Planning process group accounts for 24 of 40 processes because getting the plan right is the PM's primary responsibility — and the PMP exam reflects this weighting.
- Integrated Change Control is one of the most tested M&C processes: all changes must flow through it regardless of their source (scope, schedule, cost, risk).
- Progressive elaboration means plans are developed in detail as more information becomes available — initial estimates are refined continuously throughout planning.
- Knowledge Areas (legacy view): Integration, Scope, Schedule, Cost, Quality, Resources, Communications, Risk, Procurement, Stakeholder — still useful for grouping ITTOs for exam study.

Key Terms & Definitions

Term	Definition
Process Group	One of five logical clusters (Initiating, Planning, Executing, M&C, Closing) that group processes by when they typically occur.
Knowledge Area	PMBOK 6-era grouping of processes by subject matter (e.g., Schedule Management covers all schedule processes).
Integrated Change Control	The process of reviewing, approving, or rejecting all change requests across the entire project.
Progressive Elaboration	The iterative refinement of plans and estimates as more detail becomes available during the project.
Baseline	Approved version of a plan (scope, schedule, or cost) used as the reference point for measuring performance.
Work Performance Data	Raw observations and measurements captured during executing — the primary input to monitoring processes.

✦ EXAM TIPS

- Memorise the count by Process Group: 2 Initiating, 24 Planning, 10 Executing, 12 M&C, 1 Closing. Exam questions often test whether you know which group a process belongs to.
- When a question says 'the project manager is reviewing actual vs. planned performance,' you are in Monitoring & Controlling — specifically likely Control Scope, Control Schedule, or Control Costs.
- Do not confuse 'project phases' with 'process groups.' Process groups can repeat in every phase of a multi-phase project.
- The single Closing process (Close Project or Phase) is heavily tested. It produces lessons-learned documentation and formally releases resources — these outputs appear in distractors frequently.
- On the exam, if work has already started but something unexpected occurs, the correct action almost always routes through Perform Integrated Change Control before any other action.

✦ KNOWLEDGE CHECK

1. Sarah is two months into construction when the city asks for a new wing. Which process must she invoke before any scope change can be approved, and which process group does it belong to?
2. How many processes are in the Planning Process Group, and why is it the largest group?
3. A colleague says "We finished planning, so monitoring starts after execution ends." What is wrong with this statement?

4. Work Performance Data differs from Work Performance Information. Explain the difference and name one process that produces each.

Starting the Project — Initiate Project or Phase

Overview

Lecture 3 focuses on the two Initiating processes: Develop Project Charter and Identify Stakeholders. For the Riverside Medical Center project, these processes establish the legal and political foundation before a single brick is laid.

Develop Project Charter formalises the project's existence. The city issues a signed charter naming Sarah as PM, defining the $50M budget envelope, and authorising her to draw on city resources. Without it, Sarah has no formal authority.

Identify Stakeholders goes broader than most students expect. It is not just the construction team — it includes the city council, future patients, neighbourhood residents, the hospital board, regulatory bodies, and media. Early identification enables early engagement.

Key Concepts

- Develop Project Charter inputs: Business Case (why the hospital is needed), Benefits Management Plan (how benefits will be realised), Agreements (contracts with the city), Enterprise Environmental Factors, and Organisational Process Assets.
- The project charter documents: project purpose, measurable objectives, high-level requirements, summary milestone schedule, pre-approved budget, stakeholder list, assigned PM and authority level, and sponsor signature.
- Identify Stakeholders inputs: Project Charter (first use), Business Documents, Agreements, EEFs, OPAs — and after first iteration, Project Management Plan and project documents.
- The Stakeholder Register (output of Identify Stakeholders) captures: identification information, assessment information (influence, interest, engagement level), and classification.
- Stakeholder identification is iterative — the register is updated throughout the project as new stakeholders emerge or existing ones change roles.
- Expert Judgment is a tool in both Initiating processes — the PM consults subject matter experts, consultants, and industry groups to validate assumptions.
- The Business Case justifies the project financially and strategically. It is produced before the charter and drives the sponsor's decision to authorise.
- A Project Management Office (PMO) often provides templates and governance frameworks (OPAs) that shape how the charter and stakeholder register are formatted.
- High-level risks documented in the charter are not risk analysis — they are first-pass flags that get fully elaborated in the Plan Risk Management and Identify Risks processes later.

Key Terms & Definitions

Term	Definition
Project Charter	Output of Develop Project Charter; formally authorises the project and the PM's authority to use resources.
Business Case	Document that describes the need for the project, its expected benefits, and justification for the investment.
Stakeholder Register	Document identifying stakeholders and classifying them by interest, influence, involvement, and engagement level.
Enterprise Environmental Factors (EEF)	Conditions outside the project team's control that influence decisions: culture, market conditions, regulations, infrastructure.
Organisational Process Assets (OPA)	Plans, processes, policies, procedures, and knowledge bases from the performing organisation.
Assumption Log	Document that records assumptions and constraints identified during initiation; updated throughout the project.

✦ EXAM TIPS

- The project charter is signed by the sponsor, not the project manager. If a question asks who authorises the project, the answer is the sponsor.
- Identify Stakeholders is both an Initiating process AND is repeated throughout the project. Exam questions that say 'a new stakeholder appears mid-project' expect you to update the Stakeholder Register.
- EEFs are inputs the team cannot control; OPAs are assets the team can use. Distinguishing these two on the exam is worth several correct answers.
- The Assumption Log is created in Develop Project Charter and is one of the most underestimated outputs — it feeds directly into risk identification.
- A Benefits Management Plan is an input to the charter, not an output. Do not confuse it with the charter itself.

✦ KNOWLEDGE CHECK

1. The city's hospital board has approved funding for Riverside Medical Center. What document must now be created before Sarah can officially begin managing the project, and who must sign it?
2. Sarah discovers that a neighbourhood residents' association near the hospital site has strong opinions about construction hours. At what point should they be added to the Stakeholder Register?
3. Distinguish between an Enterprise Environmental Factor and an Organisational Process Asset. Give one example of each relevant to the hospital project.

4. What is the Assumption Log and why is it important for risk management later in the project?

Building the Master Plan — Integrate and Align Project Plans

Overview

Lecture 4 covers the integration planning processes: Develop Project Management Plan, which produces the master plan that coordinates all 10 subsidiary plans. For Riverside Medical Center, the PMP becomes the governing document that aligns the construction schedule, the medical equipment procurement timeline, and the IT infrastructure rollout.

The Project Management Plan is not a single document — it is a collection of subsidiary plans (scope, schedule, cost, quality, resource, communications, risk, procurement, stakeholder) plus baselines (scope, schedule, cost). Understanding this structure is critical for the exam.

The lecture also covers how configuration management and change management are built into the PMP itself, ensuring that any deviation from plan goes through a formal approval process rather than being handled informally.

Key Concepts

- Develop Project Management Plan is the only process whose primary input is outputs from all other planning processes — it is the integration hub of the Planning Process Group.
- The 10 subsidiary plans: Scope Management Plan, Requirements Management Plan, Schedule Management Plan, Cost Management Plan, Quality Management Plan, Resource Management Plan, Communications Management Plan, Risk Management Plan, Procurement Management Plan, Stakeholder Engagement Plan.
- Three baselines constitute the Performance Measurement Baseline (PMB): Scope Baseline (approved WBS + WBS dictionary + project scope statement), Schedule Baseline, and Cost Baseline.
- The Change Management Plan (part of the PMP) defines how changes will be submitted, evaluated, approved, and communicated — preventing scope creep.
- Configuration Management Plan (part of the PMP) defines how project documents and deliverables are versioned, stored, and controlled.
- Tailoring: PMBOK 8 emphasises that the PM must tailor the process to fit the project. Not every project needs all 10 subsidiary plans at the same depth.
- The PMP is progressively elaborated — the initial version is high-level; it gains detail as planning processes complete and baselines are set.
- Kickoff meeting marks the transition from planning to executing — it is convened after the PMP is approved and communicates the plan to the team.
- Once approved, the PMP can only be changed through Perform Integrated Change Control — this is what makes it a controlled document.

Key Terms & Definitions

Term	Definition
Project Management Plan (PMP)	The master document integrating all subsidiary plans and baselines; the approved guide for executing and controlling the project.
Scope Baseline	Approved project scope statement, WBS, and WBS dictionary used to measure scope variances.
Performance Measurement Baseline	Integrated scope-schedule-cost plan (the three baselines combined) used for Earned Value Management.
Configuration Management Plan	Defines procedures for identifying, recording, and controlling changes to project documents and deliverables.
Change Management Plan	Documents the process for submitting, reviewing, approving, or rejecting changes throughout the project.
Kickoff Meeting	Meeting held at the start of the Executing Process Group to communicate the approved plan to all project stakeholders.

✦ EXAM TIPS

- The Project Management Plan is a collection of plans and baselines — not a single calendar or Gantt chart. Exam questions try to trick you by calling a schedule 'the project plan.'
- Only Perform Integrated Change Control can approve changes to the PMP. If a question asks 'what must happen before the scope baseline can be modified,' the answer is Integrated Change Control.
- The three baselines (scope, schedule, cost) collectively form the Performance Measurement Baseline — used in Earned Value calculations.
- Know the difference between a plan and a baseline: a plan describes how something will be managed; a baseline is the approved target against which performance is measured.
- The kickoff meeting is the formal boundary between Planning and Executing — exam questions set in 'the team just received the approved plan' are in the Executing Process Group.

✦ KNOWLEDGE CHECK

1. Sarah's sponsor asks for "the project plan." Sarah provides a 200-page collection of documents. Why is this correct, and what are the three components that form the Performance Measurement Baseline?
2. Two months into construction, the hospital board wants to add a helipad. What must happen before this addition can be incorporated into the project?
3. What is the difference between the Scope Management Plan and the Scope Baseline?
4. Why is the kickoff meeting significant, and which process group does it mark the beginning of?

Managing Execution — Direct, Assure, and Learn

Overview

Lecture 5 covers the three core executing processes in Integration Management: Direct and Manage Project Work (producing deliverables), Manage Project Knowledge (capturing and sharing lessons), and Perform Quality Assurance (now renamed Manage Quality in PMBOK 6/7).

Direct and Manage Project Work is where the actual hospital gets built. Sarah authorises work packages, coordinates subcontractors, tracks deliverables, and generates Work Performance Data — the raw numbers that feed all monitoring processes.

Manage Project Knowledge ensures that insights gained during execution (what worked in the foundation pour, what didn't in the steel erection) are captured and shared so the team improves continuously and the organisation builds institutional knowledge.

Key Concepts

- Direct and Manage Project Work outputs: Deliverables, Work Performance Data, Issue Log, Change Requests, Project Management Plan updates, Project Document updates.
- Work Performance Data is raw and unanalysed — it includes actual costs incurred, % complete, defect counts, and activities started/completed. It becomes Work Performance Information only after analysis in monitoring processes.
- Issue Log records problems that have already occurred (as opposed to the Risk Register, which records potential future problems). Issues need owners and resolution dates.
- Manage Project Knowledge outputs: Lessons Learned Register and OPA updates. Lessons learned are captured throughout the project, not just at closing.
- Manage Quality focuses on auditing quality processes (process improvement) rather than inspecting deliverables (which is Control Quality in M&C).
- Quality Audit (tool in Manage Quality) independently examines whether project processes conform to organisational and project policies.
- Change Requests generated during execution must route through Perform Integrated Change Control — they may be corrective actions, preventive actions, or defect repairs.
- Corrective Action: brings future work back in line with the plan. Preventive Action: acts to reduce the probability of a future variance. Defect Repair: fixes a non-conforming component.
- The PM spends most of their time in Direct and Manage Project Work — it is the operational heartbeat of the project.

Key Terms & Definitions

Term	Definition
Deliverable	Any unique, verifiable product, result, or capability produced as part of a project process or phase.
Work Performance Data	Raw measurements from executing: costs incurred, activities completed, defects found. Not yet analysed.
Issue Log	Document recording all active issues: description, owner, due date, and resolution status.
Corrective Action	Intentional action to realign future work performance with the project management plan.
Preventive Action	Intentional action to ensure future project work performance aligns with the plan (proactive).
Lessons Learned Register	Document recording knowledge gained during the project — both positive and negative — for future use.

✦ EXAM TIPS

- Work Performance Data (raw) → Work Performance Information (analysed) → Work Performance Reports (communicated). Know this flow cold — it appears in many ITTO questions.
- The Issue Log and the Risk Register are different documents. Issues have happened; risks may happen. Exam distractors swap these frequently.
- Manage Quality ≠ Control Quality. Manage Quality audits processes; Control Quality inspects deliverables. A question about 'reviewing whether procedures are being followed' is Manage Quality.
- Change Requests can originate in any executing process and must all flow through Perform Integrated Change Control. Never implement a change without approval.
- Lessons Learned Register is updated throughout the project, not just at closing. If a question says 'the team learned something mid-project,' the answer involves updating the Lessons Learned Register.

✦ KNOWLEDGE CHECK

1. Sarah's construction team completes the hospital foundation ahead of schedule. What type of document captures this actual progress data, and what is it called at this raw stage?
2. A subcontractor installs the wrong grade of medical gas piping. Classify the PM's response: is it a Corrective Action, Preventive Action, or Defect Repair? Justify your answer.
3. What is the difference between Manage Quality and Control Quality? Give a hospital project example of each.

4. Why should lessons learned be captured throughout execution rather than only at project closing?

Monitoring Performance, Controlling Changes, and Closing Strong

Overview

Lecture 6 covers the M&C and Closing integration processes: Monitor and Control Project Work, Perform Integrated Change Control, and Close Project or Phase. These three processes keep the hospital project on track, manage deviations formally, and ultimately hand the building over to the city.

Monitor and Control Project Work is the 'dashboarding' process — it compares actuals to plan, forecasts future performance, and recommends corrective actions. For Riverside Medical Center, Sarah runs weekly performance reviews using Earned Value data.

Close Project or Phase is a single process but one of the highest-stakes on the exam. It transfers the finished hospital to the city, obtains formal acceptance, archives project records, releases team members, and updates the Lessons Learned Repository.

Key Concepts

- Monitor and Control Project Work inputs include Work Performance Information (already analysed), and it produces Work Performance Reports and Change Requests.
- Perform Integrated Change Control (PICC) evaluates every change request for impact across the triple constraint: scope, schedule, and cost. The Change Control Board (CCB) approves or rejects changes.
- PICC outputs: Approved Change Requests, Change Log updates, PMP updates, Project Document updates. Note: only APPROVED change requests proceed to Direct and Manage Project Work for implementation.
- Close Project or Phase inputs: Project Charter, PMP, Project Documents (especially Lessons Learned Register), Accepted Deliverables, Business Documents, Agreements, OPAs.
- Final Product/Service/Result Transition is an output of Close Project or Phase — the formal handover of the hospital to city operations.
- Project archives and updated OPAs ensure the organisation retains knowledge for future projects. This is the mechanism by which institutional knowledge compounds over time.
- A project can be closed prematurely (cancelled) — Close Project or Phase still applies. Document what was done, lessons learned, and formally release resources.
- The Change Control Board (CCB) is the governance body that reviews and decides on change requests. The PM may be on the CCB but rarely has unilateral approval authority.
- Work Performance Reports (output of Monitor and Control) include status reports, progress reports, forecasts, variance analysis, and earned value reports.

Key Terms & Definitions

Term	Definition
Change Control Board (CCB)	Formally chartered group responsible for reviewing, evaluating, approving, or rejecting change requests.
Change Log	Comprehensive list of changes submitted during the project, including their status (pending, approved, rejected, deferred).
Accepted Deliverables	Deliverables that have passed Control Quality inspection and been formally accepted by the customer or sponsor.
Final Product Transition	Formal handover of the completed project deliverable to the operating organisation or customer.
Work Performance Reports	Physical or electronic representation of Work Performance Information used in decision-making and communications.
Project Closure Document	Final document recording what was accomplished, lessons learned, and formal closure of all project activities.

✦ EXAM TIPS

• PICC does not implement changes — it approves them. Implementation happens in Direct and Manage Project Work. This sequence is heavily tested.

• Close Project or Phase applies even when a project is cancelled. The exam frequently presents a scenario where the project is terminated and asks what the PM should do first — always close formally.

• The difference between Accepted Deliverables (input to Close Project) and Verified Deliverables (input to Validate Scope) is a common distractor. Verified = passed quality checks; Accepted = customer sign-off.

• Monitor and Control Project Work produces Change Requests, not decisions. Decisions on those requests come from PICC.

• Lessons Learned Repository (OPA updated in Closing) vs Lessons Learned Register (project document updated throughout execution) — different names, different timing.

✦ KNOWLEDGE CHECK

1. The hospital radiology wing is over budget by 15%. Sarah submits a change request to reallocate contingency funds. Which process evaluates this request, and who has authority to approve it?

2. The city cancels the helipad component mid-project due to budget cuts. Does Sarah still need to formally close that phase? What outputs must she produce?

3. What is the difference between Verified Deliverables and Accepted Deliverables, and which comes first in the process flow?
4. Name three items Sarah must complete as part of Close Project or Phase when the hospital is handed over to city operations.

Defining the Hospital's Scope — Plan, Gather, and Define

Overview

Lecture 7 covers the three scope planning processes: Plan Scope Management, Collect Requirements, and Define Scope. Together they answer the foundational question: exactly what will be built, and how will scope decisions be made?

For Riverside Medical Center, collecting requirements means interviewing clinicians, nurses, administrators, patients, and regulatory bodies — each with different expectations. The Requirements Traceability Matrix ensures that every requirement can be traced back to its source and forward to a deliverable.

Define Scope transforms the collected requirements into a precise Project Scope Statement that draws the line between what is and what is not included in the project — the explicit exclusions are as important as the inclusions.

Key Concepts

- Plan Scope Management outputs: Scope Management Plan (how scope will be defined, validated, and controlled) and Requirements Management Plan (how requirements will be elicited, analysed, documented, and managed).
- Collect Requirements tools: Interviews, Focus Groups, Facilitated Workshops (JAD sessions), Questionnaires & Surveys, Observations (job shadowing), Prototypes, Benchmarking, Context Diagrams, Document Analysis.
- Requirements Traceability Matrix (RTM): maps each requirement to its business need, project objective, deliverable, and test case. Ensures no requirement is orphaned or gold-plated.
- Types of requirements: Business Requirements (why), Stakeholder Requirements (who needs what), Solution Requirements (what the system/product must do — functional and non-functional), Transition Requirements, Project Requirements, Quality Requirements.
- Define Scope transforms requirements into the Project Scope Statement: detailed description of deliverables, acceptance criteria, exclusions, constraints, and assumptions.
- Gold Plating: adding features beyond what was agreed — a scope management failure even if the additions seem beneficial. Always prohibited.
- Scope Creep: unauthorised scope growth without formal change control. Distinct from gold plating (internal) — scope creep often originates from external stakeholder pressure.
- Product Scope vs Project Scope: Product scope describes features and functions of the deliverable; Project scope describes the work required to deliver it. Both must be controlled.
- Expert Judgment and Data Analysis (alternatives analysis) are key tools in Define Scope for choosing between competing technical or design approaches.

Key Terms & Definitions

Term	Definition
Project Scope Statement	Output of Define Scope; describes deliverables, acceptance criteria, exclusions, constraints, and assumptions.
Requirements Traceability Matrix	Grid linking each requirement to its source, objectives, deliverables, and test cases for end-to-end traceability.
Scope Creep	Uncontrolled expansion of project scope without formal change control, schedule adjustment, or budget increase.
Gold Plating	Adding features or enhancements beyond what was agreed, without a change request — a quality and scope risk.
Product Scope	Features and functions of the deliverable (what it does).
Project Scope	Work required to deliver the product, service, or result with the specified features and functions.

✦ EXAM TIPS

- Know the difference between Collect Requirements (eliciting what stakeholders need) and Define Scope (deciding what the project will deliver). Collect comes first.
- The RTM is one of the most tested outputs in scope management. A question about 'tracing a requirement back to its business objective' points to the RTM.
- Scope creep is different from an approved scope change. Scope creep happens without formal change control; an approved scope change goes through PICC.
- Acceptance criteria are part of the Project Scope Statement — not the WBS. If a question asks 'where are the criteria for accepting a deliverable documented,' the answer is the scope statement.
- Gold plating fails the project even if the additions are technically correct, because it may consume budget/schedule and was never formally agreed with the customer.

✦ KNOWLEDGE CHECK

1. Sarah holds workshops with doctors, nurses, and city administrators to understand what the hospital must deliver. Which process is she performing, and what document captures all the gathered information?
2. A construction engineer suggests adding a rooftop garden "while they're up there anyway." No change request is filed. What problem does this create, and what is the term for it?
3. What is the Requirements Traceability Matrix and why is it important for scope control?
4. Distinguish between Product Scope and Project Scope using the hospital project as your example.

Organizing the Work — WBS, Scope Control, and Validation

Overview

Lecture 8 covers the three remaining scope processes: Create WBS, Validate Scope, and Control Scope. The WBS is the structural backbone of the entire project — every activity, resource estimate, and cost must trace back to a WBS element.

Validate Scope is a customer-facing process in the Monitoring & Controlling group that obtains formal acceptance of completed deliverables. It is often confused with Control Quality — understanding the distinction is worth several exam points.

Control Scope monitors scope performance, manages scope baseline changes through PICC, and prevents scope creep from eroding the project's value.

Key Concepts

- Create WBS decomposes the total project scope into smaller, manageable components. The lowest level is a Work Package — a piece of work that can be scheduled, cost-estimated, monitored, and controlled.
- WBS Dictionary: companion document to the WBS. Contains for each WBS element: identifier code, description, assumptions, responsible organisation, schedule milestones, required resources, cost estimate, quality requirements, acceptance criteria.
- 100% Rule: the WBS must capture 100% of the project scope — no more, no less. Any work not in the WBS is outside scope; any in-scope work not in the WBS is missing.
- Scope Baseline = Project Scope Statement + WBS + WBS Dictionary (three documents together).
- Validate Scope inputs: Verified Deliverables (from Control Quality), Requirements Documentation, RTM. Output: Accepted Deliverables and Change Requests.
- Control Scope inputs: Work Performance Data and WBS; outputs: Work Performance Information, Change Requests, Scope Baseline updates.
- Variance Analysis in Control Scope: compares actual scope performance to the scope baseline to determine the magnitude of variance and decide on corrective action.
- Rolling Wave Planning: a technique where near-term work packages are planned in detail while far-future work packages remain at a higher level (planning packages). Used in large projects like hospital construction.
- Decomposition is the primary tool for both Create WBS and Define Activities — in Create WBS it produces work packages; in Define Activities it breaks work packages into activities.

Key Terms & Definitions

Term	Definition

Work Breakdown Structure (WBS)	Hierarchical decomposition of total project scope into work packages that are manageable and deliverable-focused.
Work Package	Lowest level of the WBS — a schedulable, estimable, and assignable unit of work.
WBS Dictionary	Document defining each WBS element: code, description, resources, cost, schedule, acceptance criteria.
100% Rule	WBS must include 100% of project scope and nothing outside it — captures all deliverables and work.
Validate Scope	M&C process of obtaining formal customer acceptance of completed deliverables. Produces Accepted Deliverables.
Rolling Wave Planning	Progressive elaboration technique: plan near-term work in detail, keep future work at summary level.

✦ EXAM TIPS

- Validate Scope is performed by the customer/sponsor. Control Quality is performed by the project team. A question about 'customer sign-off' = Validate Scope.
- The 100% Rule is the most important rule for WBS creation. If a question presents a WBS scenario and one option adds deliverables not in scope, it violates the 100% Rule.
- The Scope Baseline is three documents: scope statement + WBS + WBS dictionary. If any of the three changes, the baseline changes and PICC is required.
- Validate Scope can result in Change Requests (if deliverables are not accepted). Do not assume Validate Scope always produces acceptance.
- Rolling Wave Planning is the correct answer for any question about 'planning in detail only what is known now' or 'iterative planning as the project progresses.'

✦ KNOWLEDGE CHECK

1. Sarah creates a hierarchical chart breaking the hospital project into sections: structure, MEP, IT, furnishings. What is this document, and what is the lowest level element called?
2. The city's medical director reviews the completed emergency department and formally signs off on it. Which process just occurred, and what is the key output?
3. How does the 100% Rule apply to Sarah's WBS, and what would be the consequence of a work package being omitted?
4. Explain Rolling Wave Planning and describe a situation in the hospital project where it would be appropriate.

The Three-Year Timeline — All Three Schedule Processes

Overview

Lecture 9 covers all three schedule management processes: Plan Schedule Management, and then the core processes — Define Activities, Sequence Activities, Estimate Activity Durations, Develop Schedule, and Control Schedule. The hospital's three-year timeline must be planned at the activity level before a single resource is committed.

The lecture walks through the scheduling workflow in sequence: activities are identified, dependencies are mapped using a network diagram, durations are estimated, and the Critical Path is calculated. Only then can a realistic baseline schedule be approved.

Control Schedule monitors actual progress against the baseline, analyses variances using Earned Value and Schedule Performance Index (SPI), and triggers change requests when significant deviations occur.

Key Concepts

- Define Activities: decomposes WBS work packages into schedule activities — the actual work the team will perform. Outputs: Activity List, Activity Attributes, Milestone List.
- Sequence Activities: determines the logical relationships between activities. Outputs: Project Schedule Network Diagram, Project Document updates.
- Four dependency types (Precedence Diagramming Method): Finish-to-Start (FS), Start-to-Start (SS), Finish-to-Finish (FF), Start-to-Finish (SF — rare).
- Leads and Lags: Lead = overlap (negative lag); Lag = delay (positive lag). Used to model real-world dependencies like curing time for concrete.
- Estimate Activity Durations tools: Analogous Estimating (from similar past projects), Parametric Estimating (statistical), Three-Point Estimating (PERT: (O + 4M + P)/6), Bottom-Up Estimating, Reserve Analysis.
- Develop Schedule outputs: Project Schedule (baseline once approved), Schedule Data, Schedule Baseline, Project Calendars, Change Requests.
- Critical Path Method (CPM): identifies the longest sequence of activities. Total Float = zero on critical path. Schedule can only be compressed by shortening critical path activities.
- Schedule Compression techniques: Crashing (adding resources to critical path activities, increases cost) and Fast Tracking (parallelising activities, increases risk).
- Control Schedule: SPI = EV/PV. SPI < 1 = behind schedule. SPI > 1 = ahead of schedule.

Key Terms & Definitions

Term	Definition

Critical Path	Longest sequence of dependent activities determining the project's minimum duration; activities on it have zero total float.
Total Float	Amount of time an activity can be delayed without delaying the project end date.
Crashing	Schedule compression by adding resources to critical path activities to shorten duration — increases cost.
Fast Tracking	Schedule compression by running activities in parallel that were originally sequential — increases risk.
Three-Point Estimate (PERT)	Duration estimate using Optimistic, Most Likely, and Pessimistic: (O + 4M + P) ÷ 6.
Schedule Performance Index (SPI)	EV ÷ PV. Ratio measuring schedule efficiency. <1 = behind schedule, >1 = ahead.

✦ EXAM TIPS

- CPM questions appear on nearly every PMP exam. Practice calculating early start, early finish, late start, late finish, and float. Critical path = zero float path.
- Crashing vs Fast Tracking: if the question asks about compressing schedule by adding resources → Crashing. If activities are run in parallel → Fast Tracking.
- SPI < 1 is behind schedule; CPI < 1 is over budget. Know both ratios and what they signal.
- PERT formula: (O + 4M + P) / 6 for expected duration; standard deviation = (P - O) / 6. Both appear in quantitative exam questions.
- Leads are used to overlap activities (e.g., start testing before all coding is complete). Lags are delays. Do not confuse these with float.

✦ KNOWLEDGE CHECK

1. The hospital project has the following activity chain: Foundation (20 days) → Structure (45 days) → MEP rough-in (30 days) → Finishes (25 days). What is the critical path duration, and what does zero total float mean for these activities?
2. Sarah is 8 weeks behind on the schedule baseline. Her sponsor demands she recover 4 weeks. She can either add a second construction crew (cost: $200K) or run MEP and finishes simultaneously. Which technique is each option, and what is the primary risk of each?
3. A concrete pour requires 7 days of curing before the next activity can start. How is this modelled in the schedule network, and what is the scheduling term for this delay?
4. Calculate the PERT expected duration for an activity with Optimistic = 4 days, Most Likely = 7 days, Pessimistic = 16 days.

The Fifty Million Dollar Budget — All Four Finance Processes

Overview

Lecture 10 covers all four cost management processes: Plan Cost Management, Estimate Costs, Determine Budget, and Control Costs. The $50 million hospital budget is not a single figure — it is built from the ground up through activity-level estimates, aggregated into a cost baseline, and protected by management reserves.

Estimate Costs links directly to the WBS: every work package gets a cost estimate. The estimates roll up through Determine Budget to form the cost baseline, which is the S-curve against which actual spend is measured using Earned Value Management.

Control Costs is where Earned Value Management (EVM) lives. The exam dedicates significant question weight to EVM formulas — SV, CV, SPI, CPI, EAC, ETC, TCPI. Understanding what each metric reveals about project health is non-negotiable.

Key Concepts

- Plan Cost Management outputs: Cost Management Plan (defines cost units, precision, control thresholds, Earned Value rules, and reporting formats).
- Estimate Costs tools: Analogous, Parametric, Bottom-Up, Three-Point (PERT), Reserve Analysis (contingency). Key output: Cost Estimates and Basis of Estimates.
- Determine Budget: aggregates individual cost estimates plus contingency reserves to establish the cost baseline. Management Reserves are added on top of the baseline.
- Cost Baseline (S-curve) = authorised budget excluding Management Reserves. Management Reserves are held by the sponsor/organisation for unknown unknowns.
- Contingency Reserves are for known unknowns (identified risks). They are inside the cost baseline. Management Reserves are for unknown unknowns and are outside the baseline.
- EVM Core: PV (Planned Value) = budgeted cost of work scheduled; EV (Earned Value) = budgeted cost of work performed; AC (Actual Cost) = actual cost of work performed.
- Variances: SV = EV - PV (negative = behind schedule); CV = EV - AC (negative = over budget).
- Performance Indices: SPI = EV/PV; CPI = EV/AC. CPI is the most reliable predictor of final project cost.
- Forecasting: EAC = BAC/CPI (most common); ETC = EAC - AC; TCPI = (BAC - EV) / (BAC - AC).

Key Terms & Definitions

Term	Definition

Planned Value (PV)	Authorised budget assigned to scheduled work — what you planned to spend by this point.
Earned Value (EV)	Authorised budget for work actually completed — the value of work accomplished.
Actual Cost (AC)	Realised cost incurred for work performed during a specific time period.
Cost Performance Index (CPI)	EV ÷ AC. <1 = over budget; >1 = under budget. Most reliable forecast predictor.
Estimate at Completion (EAC)	Expected total cost of the project: most common formula = BAC ÷ CPI.
Contingency Reserve	Budget set aside for identified risks (known unknowns); included in the cost baseline.

✦ EXAM TIPS

- CPI < 1 means you are getting less value than you are spending — over budget. CV negative = over budget. SV negative = behind schedule. Memorise all four signs.
- EAC = BAC / CPI is the most commonly tested forecasting formula. Use it when current performance is expected to continue.
- Contingency Reserves (for known unknowns, inside baseline) vs Management Reserves (for unknown unknowns, outside baseline) — a classic exam distinction.
- TCPI > 1 means the project needs to be MORE efficient going forward than it has been — a warning sign. TCPI < 1 means it can afford to be less efficient.
- The cost baseline is the S-curve — it reflects cumulative planned spend over time. Changes to it require PICC approval.

✦ KNOWLEDGE CHECK

1. At month 18 of the hospital project, PV = $22M, EV = $20M, AC = $23M. Calculate SV, CV, SPI, and CPI. Is the project ahead or behind schedule, and over or under budget?
2. The hospital's BAC is $50M. After month 18 (from question 1), what is the EAC assuming current efficiency continues? What does this mean for the project?
3. Explain the difference between Contingency Reserve and Management Reserve. Who can authorise use of each?
4. What does a CPI of 0.87 tell Sarah about the hospital project, and what action should she take?

The Hospital's Stakeholders — All Seven Processes

Overview

Lecture 11 covers all seven stakeholder management processes: Identify Stakeholders, Plan Stakeholder Engagement, Manage Stakeholder Engagement, and Monitor Stakeholder Engagement — plus the three communications processes that run in parallel: Plan Communications Management, Manage Communications, and Monitor Communications.

For Riverside Medical Center, stakeholders span from the city council to future patients. The Stakeholder Engagement Assessment Matrix maps each stakeholder's current engagement level (Unaware, Resistant, Neutral, Supportive, Leading) against the desired level, revealing engagement gaps that the PM must close.

Communications management is often the difference between a technically successful project that fails stakeholder expectations and one that succeeds on all dimensions. The communications plan defines who gets what information, when, in what format, through which channel.

Key Concepts

- Stakeholder Engagement Assessment Matrix: columns = engagement levels (Unaware, Resistant, Neutral, Supportive, Leading). 'C' marks current; 'D' marks desired. The PM must move each stakeholder from C to D.
- Plan Stakeholder Engagement outputs: Stakeholder Engagement Plan (how to increase support and reduce resistance). This is separate from the Communications Management Plan.
- Manage Stakeholder Engagement (Executing): implements the engagement plan through negotiation, conflict resolution, and facilitation. Output: Change Requests and updated Stakeholder Register.
- Monitor Stakeholder Engagement (M&C): tracks whether engagement strategies are working. Uses Work Performance Information to identify resistant or disengaged stakeholders.
- Communications Requirements Analysis: determines the total information needs of stakeholders. Formula: N(N-1)/2 communication channels for N stakeholders.
- Communication methods: Interactive (real-time, two-way, e.g., meetings), Push (sent without request, e.g., email, reports), Pull (receiver accesses when needed, e.g., intranet, shared drives).
- Noise in communication: anything that distorts the message — language barriers, information overload, cultural differences, technical jargon. Acknowledged in the Sender-Receiver model.
- Plan Communications Management outputs: Communications Management Plan (who, what, when, how, frequency, responsible person for each communication).

- Manage Communications outputs: Project Communications (the actual communications artefacts), OPA updates, PMP updates.

Key Terms & Definitions

Term	Definition
Stakeholder Engagement Plan	Document describing strategies and actions to engage stakeholders effectively throughout the project lifecycle.
Engagement Assessment Matrix	Grid mapping stakeholder current vs desired engagement levels (Unaware/Resistant/Neutral/Supportive/Leading).
Communications Management Plan	Defines what information each stakeholder receives, by whom, via which channel, and at what frequency.
Interactive Communication	Real-time, two-way exchange between parties (meetings, phone calls, video conferences).
Push Communication	One-way communication sent to recipients without a request (emails, reports, press releases).
Pull Communication	Information available for stakeholders to access on demand (intranet portals, shared document repositories).

✦ EXAM TIPS

• The Stakeholder Engagement Assessment Matrix is the tool for stakeholder analysis. Know the five levels and what the gap between C and D means.

• N(N-1)/2 communication channels formula: if there are 10 stakeholders, there are 45 channels. Adding one person to a team adds significant complexity.

• Manage Stakeholder Engagement is Executing; Monitor Stakeholder Engagement is M&C. If the question says 'tracking whether engagement is working,' it's Monitor.

• Communications plan questions often test which method (interactive, push, pull) is most appropriate. Choose interactive for sensitive topics; push for routine updates; pull for large reference documents.

• Noise (barriers to communication) is a classic exam topic. Identify the noise and mitigate it — do not just send the message again.

✦ KNOWLEDGE CHECK

1. Sarah's team has 12 members and 8 external stakeholders. How many communication channels exist? What does this number tell her about communications planning?

2. The neighbourhood residents' association is marked as Resistant (C) but Sarah needs them to be Supportive (D). Which plan addresses this gap, and what process manages its execution?
3. A critical status report is being distributed. Is this Push or Pull communication? When would Pull communication be more appropriate?
4. What is the Stakeholder Engagement Assessment Matrix and how does it guide the PM's stakeholder strategy?

Building the Team — All Five Resources Processes

Overview

Lecture 12 covers all five resource management processes: Plan Resource Management, Estimate Activity Resources, Acquire Resources, Develop Team, and Manage Team. Plus the single Control Resources (M&C) process — six total across the resource management knowledge area.

For Riverside Medical Center, resources include construction workers, specialist engineers, medical equipment technicians, IT staff, and project management staff. Acquiring the right people at the right time, keeping them motivated, and resolving conflicts are Sarah's daily responsibilities during execution.

The lecture distinguishes between physical resources (materials, equipment, facilities) and team resources (people). Both are managed through resource management processes, but team development requires leadership skills that are tested heavily on the PMP exam.

Key Concepts

- Plan Resource Management outputs: Resource Management Plan (how resources will be estimated, acquired, managed, controlled, and released) and Team Charter (establishes team values, agreements, and operating guidelines).
- Estimate Activity Resources: determines the type and quantity of materials, people, equipment, and supplies needed. Output: Resource Requirements, Resource Breakdown Structure (RBS), Basis of Estimates.
- Acquire Resources: obtains team members and physical resources. Outputs include Physical Resource Assignments, Project Team Assignments, Resource Calendars.
- Responsibility Assignment Matrix (RAM) / RACI: maps project roles to WBS work packages. RACI = Responsible (does the work), Accountable (owns the outcome), Consulted (provides input), Informed (kept in the loop).
- Develop Team tools: Co-location, Virtual Teams, Communication Technology, Interpersonal & Team Skills, Recognition & Rewards, Training, Individual & Team Assessments, Ground Rules.
- Tuckman's Team Development Model: Forming → Storming → Norming → Performing → Adjourning. PM adapts leadership style to each stage.
- Conflict Resolution techniques: Collaborate/Problem Solve (win-win, best long-term), Compromise/Reconcile (both parties give up something), Force/Direct (win-lose), Smooth/Accommodate (keep peace), Withdraw/Avoid (postpone).
- Manage Team: tracks performance, provides feedback, resolves issues, manages changes to optimise project performance. Output: Change Requests and Team Performance Assessments.
- Control Resources: monitors planned vs actual resource utilisation and takes corrective action when variances occur.

Key Terms & Definitions

Term	Definition
RACI Matrix	Responsibility assignment tool: Responsible (does work), Accountable (owns outcome), Consulted (input), Informed (notified).
Resource Breakdown Structure	Hierarchical representation of resources by category and type used on the project.
Team Charter	Document establishing team values, agreements, ground rules, and operating guidelines for project work.
Tuckman Model	Five-stage team development: Forming, Storming, Norming, Performing, Adjourning.
Collaborate/Problem Solve	Conflict resolution technique seeking a win-win solution by incorporating multiple viewpoints; preferred approach.
Resource Calendar	Documents availability (days/hours) and capacity of specific resources required for project activities.

✦ EXAM TIPS

- Collaborate/Problem Solve is always the preferred conflict resolution technique on the exam unless a question specifically makes it impractical.
- Tuckman's model: Storming (conflict) is normal — a PM should not try to suppress it, but help the team work through it toward Norming.
- The RACI matrix ensures every work package has exactly one Accountable person — no work package should have two 'A' entries.
- Pre-assignment, Negotiation, Acquisition, and Virtual Teams are tools for Acquire Resources. Know which tool applies when internal resources are unavailable.
- Team Performance Assessments (output of Develop Team) and Individual Performance Appraisals (done in Manage Team) are different. The team-level assessment feeds Develop Team; individual appraisals inform HR actions.

✦ KNOWLEDGE CHECK

1. Sarah is managing a hospital IT implementation team that is in constant conflict over technical decisions. According to Tuckman's model, what stage is the team in, and what should Sarah do?
2. The hospital's MEP subcontractor is over-allocated for weeks 14–18. Which process addresses this, and what two techniques can resolve the resource conflict?

3. A work package for installing surgical lighting has multiple engineers who say they are "responsible." What RACI role is likely missing, and what is its definition?
4. What is the difference between Develop Team and Manage Team? Give one output of each.

Managing Risk — All Six Risk Processes on the Hospital Project

Overview

Lecture 13 covers all six risk management processes: Plan Risk Management, Identify Risks, Perform Qualitative Risk Analysis, Perform Quantitative Risk Analysis, Plan Risk Responses, and Implement Risk Responses — plus Monitor Risks (M&C) as the seventh process.

For Riverside Medical Center, risks include regulatory approval delays, subcontractor failures, ground contamination, medical equipment supply chain issues, and budget overruns on specialist systems. The risk management processes transform these threats into manageable items with owners, responses, and contingency plans.

The lecture emphasises that risk management is proactive and iterative. The Risk Register grows throughout the project as new risks are identified and residual risks from implemented responses are assessed.

Key Concepts

- Plan Risk Management outputs: Risk Management Plan (defines risk methodology, categories, probability/impact scales, risk tolerance, reporting formats, and tracking).
- Identify Risks tools: Brainstorming, Checklists, Interviews, Root Cause Analysis, SWOT Analysis, Prompt Lists, Document Analysis. Output: Risk Register (initial), Risk Report.
- Risk Register contents: risk ID, category, description, root causes, potential responses, risk owner, probability, impact, risk score, risk response, residual risks, contingency plans.
- Qualitative Risk Analysis: prioritises risks by multiplying Probability × Impact. Output: updated Risk Register with priority rankings. Input to planning responses.
- Quantitative Risk Analysis: numerically analyses combined effect of risks on project objectives. Tools: Monte Carlo Simulation, Decision Tree Analysis, Sensitivity Analysis (Tornado Diagram).
- Risk Response Strategies for Threats: Avoid (eliminate the risk), Transfer (shift impact to third party, e.g., insurance), Mitigate (reduce probability or impact), Accept (acknowledge and plan contingency or do nothing).
- Risk Response Strategies for Opportunities: Exploit (ensure opportunity occurs), Share (allocate to party best positioned to capture it), Enhance (increase probability/impact), Accept.
- Secondary Risk: a new risk created by implementing a risk response. Must be identified and analysed.
- Residual Risk: remaining risk after a response has been implemented. May require contingency reserve.
- Monitor Risks: tracks identified risks, identifies new risks, evaluates risk process effectiveness. Uses Risk Audit as a tool.

Key Terms & Definitions

Term	Definition
Risk Register	Document containing identified risks, their analysis, prioritisation, owners, and planned responses.
Probability × Impact Matrix	Tool for Qualitative Risk Analysis that scores and prioritises risks by multiplying probability rating by impact rating.
Monte Carlo Simulation	Quantitative tool that runs thousands of project scenarios to determine probability distributions for cost and schedule outcomes.
Risk Mitigation	Response strategy that reduces the probability or impact of a threat to an acceptable threshold.
Risk Transfer	Shifts the financial impact of a risk to a third party (e.g., insurance, fixed-price contracts).
Residual Risk	Risk that remains after a response has been implemented; may require contingency reserve.

✦ EXAM TIPS

• Know all four threat responses (Avoid, Transfer, Mitigate, Accept) and all four opportunity responses (Exploit, Share, Enhance, Accept). Exam questions present a scenario and ask which strategy applies.

• Transfer does not eliminate the risk — it shifts the financial consequence. The risk still exists. Insurance is the classic example.

• Secondary risks arise from implementing a risk response — e.g., bringing in a specialist contractor to mitigate a technical risk introduces a new subcontractor management risk.

• Qualitative Analysis comes before Quantitative Analysis. Not all projects require Quantitative Analysis (it is more resource-intensive).

• Monitor Risks is the M&C counterpart to all planning risk processes. A question about 'a new risk appeared during execution' → update the Risk Register via Monitor Risks.

✦ KNOWLEDGE CHECK

1. Sarah identifies a risk: ground contamination may be discovered during excavation, delaying the project by 3 months. She decides to purchase environmental insurance. Which risk response strategy is this?

2. After implementing a mitigation strategy (using a specialist drilling technique for contaminated ground), Sarah identifies a new risk: the specialist firm may be unavailable. What is this new risk called?

3. Rank the following risk response strategies for threats from most to least aggressive in terms of eliminating the risk: Accept, Transfer, Mitigate, Avoid.
4. Explain the difference between Qualitative and Quantitative Risk Analysis. When would Sarah use each on the hospital project?

The Tools That Make It All Work — From Estimating to Analysis

Overview

Lecture 14 examines the key tools and techniques that cut across multiple processes — Earned Value Management, estimating techniques, data analysis tools, decision-making methods, and interpersonal skills. Understanding these tools in depth is what separates candidates who score in the 80th percentile from those who merely pass.

The lecture revisits estimating through the lens of the entire project: how analogous, parametric, three-point, and bottom-up estimates are used in different processes and at different levels of detail. The Basis of Estimates document justifies every estimate.

Data analysis tools — Variance Analysis, Trend Analysis, Root Cause Analysis, Regression Analysis, and the Tornado Diagram — are the analytical engine of monitoring and controlling. Each has a specific use case and appears in multiple process ITTOs.

Key Concepts

- Analogous Estimating: uses actual values from similar past projects. Fast and low-cost but less accurate. Used when limited detail is available (early planning).
- Parametric Estimating: statistical relationship between historical data and variables (e.g., cost per square metre × floor area). More accurate than analogous if the parameter is reliable.
- Bottom-Up Estimating: most accurate but time-intensive. Estimate at the activity level and aggregate up. Requires a complete WBS.
- Three-Point Estimating (PERT): accounts for estimation uncertainty. Expected = (O + 4M + P)/6. Standard Deviation = (P - O)/6. Used when there is significant uncertainty.
- Earned Value Analysis: integrates scope, schedule, and cost to provide an objective performance measure. PV, EV, AC, SV, CV, SPI, CPI, EAC, ETC, TCPI.
- Variance Analysis: compares planned vs actual (SV, CV). Trend Analysis: examines whether performance is improving or deteriorating over time.
- Root Cause Analysis (RCA): identifies the fundamental cause of a defect or variance, not just the symptom. Supports corrective action that prevents recurrence.
- Tornado Diagram: horizontal bar chart used in sensitivity analysis (Quantitative Risk Analysis) to show which risks have the greatest impact on project objectives. The widest bar = highest sensitivity.
- Decision Tree Analysis: evaluates alternative strategies under uncertainty by multiplying the probability of each outcome by its impact (Expected Monetary Value — EMV). Choose the highest positive EMV.

Key Terms & Definitions

Term	Definition
Basis of Estimates	Document explaining the assumptions, constraints, and methods used to derive cost or duration estimates.
Expected Monetary Value (EMV)	Probability × Impact for a risk or decision node; used in Decision Tree Analysis.
Tornado Diagram	Sensitivity analysis bar chart showing risks ranked by their potential impact on project outcomes — widest bar = highest impact.
Root Cause Analysis	Technique for identifying the fundamental cause of a problem or defect, enabling corrective action at the source.
Trend Analysis	Examines project performance results over time to determine whether performance is improving or deteriorating.
Regression Analysis	Statistical method that examines the relationship between variables to make predictions or identify patterns.

✦ EXAM TIPS

- EMV = Probability × Impact. In a Decision Tree, multiply each branch probability by its value and sum to get the EMV of each decision. Choose the highest positive EMV.
- The Tornado Diagram does not show the probability of risks — it shows which risks, if they occur, would have the greatest impact. Do not confuse with the P×I matrix.
- Bottom-Up Estimating is most accurate but requires a complete WBS. If a question says 'the WBS is not yet available,' bottom-up is not an appropriate tool.
- Trend Analysis answers 'where are we headed?' Variance Analysis answers 'where are we now vs plan?' Both are used in Monitor and Control processes.
- Root Cause Analysis produces corrective actions that prevent recurrence — not just fixes that address symptoms. Exam questions test whether you address root causes or symptoms.

✦ KNOWLEDGE CHECK

1. A risk on the hospital project has a 40% probability of occurring and would cost $500,000 if it does. Calculate the EMV and explain how this figure should be used in the project budget.

2. Sarah needs a quick cost estimate for a new ward in month 3 of the project when little detail is available. The organisation built a similar ward 2 years ago for $2.8M. Which estimating technique should she use?

3. The hospital IT network install is consistently running 15% over budget each reporting period. Which analysis technique should Sarah use to predict future performance, and what metric supports it?

4. Explain the Tornado Diagram: what does it show, which process uses it, and what does the widest bar represent?

Sarah's Hospital Is Open — Your Exam Mastery Begins

Overview

The final lecture in Section 6 brings the Riverside Medical Center project to completion. The hospital receives its occupancy permit, passes final inspections, and is formally handed over to city health services. This lecture consolidates all 40 processes into an integrated project narrative.

Students see how the outputs of earlier processes became inputs to later ones — the project charter fed stakeholder identification, the WBS drove activity definition, activity estimates informed the schedule and cost baselines, and risk responses protected those baselines throughout execution.

The lecture closes with exam mastery guidance: how to approach ITTO questions systematically, how to use the case-study mental model in the exam room, and the three most common failure modes on PMP exam questions about the 40 processes.

Key Concepts

- Process integration is the meta-lesson of Section 6: no process exists in isolation. Every output feeds at least one downstream input. Understanding these connections is more valuable than memorising ITTOs in isolation.
- The five Process Groups form a cycle, not just a sequence. In multi-phase projects, all five groups repeat in each phase. Lessons learned from Phase 1 closing feed Phase 2 initiating.
- Accepted Deliverables (from Validate Scope) flow into Close Project or Phase. Final Product/Service/Result Transition is the tangible output — the hospital handed over and operational.
- Lessons Learned Repository (updated at closing) is an OPA — it becomes available to future projects. This is how organisations build project management maturity over time.
- The three exam failure modes: (1) confusing Executing processes with M&C processes (doing vs monitoring), (2) skipping change control (implementing changes without PICC), (3) treating Closing as optional.
- Agile and hybrid contexts still use the 40 process principles — but tailored. Iterative projects run planning and executing in cycles (sprints), with monitoring after each iteration.
- The Process Domain (41% of PMP exam ECO) maps directly to the 40 processes. Section 6 content therefore covers a significant portion of exam questions.
- A project manager's primary job throughout the project lifecycle is to protect the baselines while delivering value — every process serves this goal.
- Project Success = Scope delivered + within Schedule + within Budget + Quality met + Stakeholders satisfied + Benefits realised. All 40 processes contribute to at least one of these dimensions.

Key Terms & Definitions

Term	Definition
Final Product Transition	Formal handover of the completed project deliverable to the customer or operating organisation.
Project Management Maturity	Organisation's capability to manage projects consistently and improve over time; built through lessons learned and OPA updates.
Process Tailoring	Adapting process rigor, tools, and documentation to the specific context, size, complexity, and risk of a project.
Benefits Realisation	The point at which the project's intended business benefits are achieved, often after project closure.
Lessons Learned Repository	OPA storing lessons from completed projects, accessible to future project teams for planning and risk management.
Process Domain	One of the three ECO domains (41% of PMP exam); covers the mechanics of managing a project from initiation to closure.

✦ EXAM TIPS

- Section 6 content (40 processes) maps to the Process Domain — 41% of the exam. Master this section and you have addressed the largest single exam domain.
- When stuck on an exam question, ask: which process group is the scenario in? What process produces the output the question is asking about? Work backwards from outputs.
- Any exam scenario where a change is needed: the answer almost always routes through PICC before implementation, regardless of change type or urgency.
- Close Project or Phase is not optional and is not just administration. It produces formal acceptance, lessons learned, and releases resources — all testable outputs.
- Hybrid project questions test whether you can apply traditional process thinking within an iterative context. Baselines still exist; change control still applies; stakeholder engagement still matters.

✦ KNOWLEDGE CHECK

1. The Riverside Medical Center passes its final inspection. City Health Services signs the acceptance document. Which process has just completed, and what is the formal name of this output?
2. List the five Process Groups in order and state which is the largest (by number of processes). Why is that group the largest?
3. A new PMP candidate says "I don't need to learn ITTOs — I'll just learn concepts." Why is this strategy risky, and what approach does this section recommend?

4. How does the Lessons Learned Repository support future projects, and which process produces the final update to this repository?

Appendix

Process Group × Knowledge Area Master Map

Process Group	Knowledge Area	Processes
Initiating	Integration	Develop Project Charter
Initiating	Stakeholder	Identify Stakeholders
Planning	Integration	Develop Project Management Plan
Planning	Scope	Plan Scope Mgmt; Collect Requirements; Define Scope; Create WBS
Planning	Schedule	Plan Schedule Mgmt; Define Activities; Sequence Activities; Estimate Durations; Develop Schedule
Planning	Cost	Plan Cost Mgmt; Estimate Costs; Determine Budget
Planning	Quality	Plan Quality Management
Planning	Resource	Plan Resource Mgmt; Estimate Activity Resources
Planning	Communications	Plan Communications Management
Planning	Risk	Plan Risk Mgmt; Identify Risks; Qualitative Analysis; Quantitative Analysis; Plan Risk Responses
Planning	Procurement	Plan Procurement Management
Planning	Stakeholder	Plan Stakeholder Engagement
Executing	Integration	Direct & Manage Work; Manage Project Knowledge
Executing	Quality	Manage Quality
Executing	Resource	Acquire Resources; Develop Team; Manage Team
Executing	Communications	Manage Communications
Executing	Risk	Implement Risk Responses
Executing	Procurement	Conduct Procurements
Executing	Stakeholder	Manage Stakeholder Engagement
M&C	Integration	Monitor & Control Work; Perform Integrated Change Control
M&C	Scope	Validate Scope; Control Scope
M&C	Schedule	Control Schedule
M&C	Cost	Control Costs
M&C	Quality	Control Quality
M&C	Resource	Control Resources
M&C	Communications	Monitor Communications
M&C	Risk	Monitor Risks

M&C	Procurement	Control Procurements
M&C	Stakeholder	Monitor Stakeholder Engagement
Closing	Integration	Close Project or Phase

Exam Cheat Sheet — Key Formulas & Rules

- CPI = EV/AC | SPI = EV/PV | SV = EV−PV | CV = EV−AC
- EAC = BAC/CPI | ETC = EAC−AC | TCPI = (BAC−EV)/(BAC−AC)
- PERT Expected Duration = (O + 4M + P) / 6 | Std Dev = (P−O)/6
- Communication Channels = N(N−1)/2
- Critical Path = longest path; Total Float = 0 on CP
- Crashing = add resources (↑ cost) | Fast Tracking = parallelise (↑ risk)
- Risk Threats: Avoid > Transfer > Mitigate > Accept
- Risk Opportunities: Exploit > Share > Enhance > Accept
- Tuckman: Forming → Storming → Norming → Performing → Adjourning
- Scope Baseline = Scope Statement + WBS + WBS Dictionary
- PMB = Scope Baseline + Schedule Baseline + Cost Baseline
- Contingency Reserve = known unknowns (inside baseline); Management Reserve = unknown unknowns (outside baseline)

Most Commonly Tested Processes

- Develop Project Charter (Initiating — defines PM authority)
- Perform Integrated Change Control (all changes must flow through it)
- Create WBS (100% Rule — scope baseline anchor)
- Develop Schedule (Critical Path Method + Float calculation)
- Control Costs (Earned Value — CPI, SPI, EAC, TCPI)
- Identify Risks + Plan Risk Responses (ATMA threats / ESEA opportunities)
- Validate Scope (customer acceptance — not quality inspection)
- Close Project or Phase (always required, even on cancelled projects)